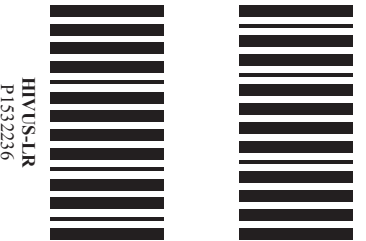


Pharmaco position may change as per Supplier's m/c requirement & additional small pharma code may appear on the front / back panel



D3 PROPOSED PACKAGE INSERT

HIVUS - LR 100/25 [Lopinavir 100mg and Ritonavir 25mg Film Coated Tablets USP]

HIVUS - LR 200/50 [Lopinavir 200mg and Ritonavir 50mg Film Coated Tablets USP]

Description

HIVUS - LR 100/25 [Lopinavir 100mg and Ritonavir 25mg Film Coated Tablets USP] are Light Yellow Film-Coated Capsule Shaped tablets debossed with 'D91' on one side and plain on the other side.

Each film coated tablet contains Lopinavir USP 100 mg and Ritonavir USP 25 mg

Coloring Agent: Titanium Dioxide, Ferric oxide yellow

HIVUS - LR 200/50 [Lopinavir 200mg and Ritonavir 50mg Film Coated Tablets USP] are Light Yellow Film-Coated tablets with ovaloid shape debossed with 'X' on one side and '28' on other side.

Each film coated Tablet contains Lopinavir USP 200 mg and Ritonavir USP 50 mg.

Coloring Agent: Titanium Dioxide, Ferric oxide yellow

Pharmacodynamics

Microbiology

Mechanism of action

Lopinavir, an inhibitor of the HIV-1 and HIV-2 proteases, prevents cleavage of the gag-pol polyprotein, resulting in the production of immature, non-infectious virions.

Antiviral activity in vitro

The *in vitro* antiviral activity of lopinavir against laboratory HIV strains and clinical HIV isolates was evaluated in acutely infected lymphoblastic cell lines and peripheral blood lymphocytes, respectively.

In the absence of human serum, the mean 50% effective concentration (EC₅₀) of lopinavir against five different HIV-1 laboratory strains ranged from 10 to 27 nM (0.006 to 0.017 mcg/mL, 1 mcg/mL equals 1.6 micro M) and ranged from 4 to 11 nM (0.003 to 0.007 mcg/mL) against several HIV-1 clinical isolates (n equals 6). In the presence of 50% human serum, the mean EC₅₀ of Lopinavir against these five laboratory strains ranged from 65 to 289 nM (0.04 to 0.18 mcg/mL), representing a 7-to 11-fold attenuation. Combination drug activity studies with lopinavir and other protease inhibitors or reverse transcriptase inhibitors have not been completed.

Resistance

HIV-1 isolates with reduced susceptibility to lopinavir have been selected in vitro. The presence of ritonavir does not appear to influence the selection of lopinavir-resistant viruses *in-vitro*.

The selection of resistance to Lopinavir and Ritonavir Film coated tablets in antiretroviral treatment naive patients has not yet been characterized. In a Phase III study of 653 antiretroviral treatment naive patients (Study 863), plasma viral isolates from each patient on treatment with plasma HIV greater than 400 copies/mL at week 24, 32, 40 and/or 48 were analyzed. No evidence of genotypic or phenotypic resistance to Lopinavir and Ritonavir Film coated tablets was observed in 37 evaluable Lopinavir and Ritonavir Film coated tablets-treated patients (0%). Evidence of genotypic resistance to nevirapine, defined as the presence of the D30N and/or L90M mutation in HIV protease, was observed in 25/76 (33%) of evaluable nevirapine-treated patients. The selection of resistance to Lopinavir and Ritonavir Film coated tablets in antiretroviral treatment naive pediatric patients (Study 940) appears to be consistent with that seen in adult patients (Study 863).

Resistance to Lopinavir and Ritonavir Film coated tablets has been noted to emerge in patients treated with other protease inhibitors prior to Lopinavir and Ritonavir Film coated tablets therapy. In Phase II studies of 227 antiretroviral treatment naive and protease inhibitor-naïve patients, isolates from 4 of 23 patients with quantifiable (greater than 400 copies/mL) viral RNA following treatment with Lopinavir and Ritonavir Film coated tablets for 12 to 100 weeks displayed significantly reduced susceptibility to lopinavir compared to the corresponding baseline viral isolates. Three of these patients had previously received treatment with a single protease inhibitor (nadiravir, indinavir, or saquinavir) and one patient had received treatment with multiple protease inhibitors (indinavir, saquinavir and ritonavir). All four of these patients had at least four mutations associated with protease inhibitor resistance immediately prior to Lopinavir and Ritonavir Film coated tablets therapy. Following viral rebound, isolates from these patients all contained additional mutations, some of which are recognized to be associated with protease inhibitor resistance. However, there are insufficient data at this time to identify lopinavir-associated mutational patterns in isolates from patients on Lopinavir and Ritonavir Film coated tablets therapy. The assessment of these mutational patterns is under study.

Cross-Resistance

Preclinical Studies

Varying degrees of cross-resistance have been observed among protease inhibitors. The *in-vitro* activity of lopinavir against clinical isolates from patients previously treated with a single protease inhibitor was determined. Isolates that displayed greater than 4-fold reduced susceptibility to nevirapine (n=13) and saquinavir (n=4), displayed less than 4-fold reduced susceptibility to lopinavir.

Isolates with greater than 4-fold reduced susceptibility to indinavir (n=16) and ritonavir (n=3) displayed a mean of 5.7- and 8.3-fold reduced susceptibility to lopinavir, respectively. Isolates from patients previously treated with two or more protease inhibitors showed greater reductions in susceptibility to lopinavir as described in the clinical section below (see Clinical Studies: Antiviral Activity of Lopinavir and Ritonavir Film coated tablets in Patients With Previous Protease Inhibitor Therapy).

Cross-Resistance During lopinavir/ritonavir Therapy

Little information is available on the cross-resistance of viruses selected during therapy with Lopinavir and Ritonavir Film coated tablets. Isolates from four patients previously treated with one or more protease inhibitors that developed increased lopinavir phenotypic resistance during Lopinavir and Ritonavir Film coated tablets therapy either remained cross-resistant or developed cross-resistance to ritonavir, indinavir, and nevirapine. All rebound viruses either remained fully sensitive or demonstrated modestly reduced susceptibility to amprenavir (up to 8.5-fold concurrent with 99-fold resistance to lopinavir). The rebound isolates from the two subjects with no prior saquinavir treatment remained fully sensitive to saquinavir.

Genotypic correlates of reduced virologic response in antiretroviral-experienced patients initiating a Lopinavir and Ritonavir Film coated tablets-based combination regimen.

Virologic response to Lopinavir and Ritonavir Film coated tablets has been shown to be affected by the presence of three or more of the following amino acid substitutions in protease at baseline: L101F/R/V, K20M/N/R, L24I, L33F, M36I, I47V, G48V, I54I/T/V, V82A/C/F, S/T, and I84V. **Table 6** shows the 48-week virologic response (HIV RNA <400 copies/mL) according to the number of the above protease inhibitor resistance mutations at baseline in studies 888 and 765 and study 957 (see below).

Table 6 Virologic Response (HIV RNA <400 copies/mL) at Week 48 by Baseline Lopinavir and Ritonavir Film coated Tablets Susceptibility and by Number of Protease Substitutions Associated with Reduced Response to Lopinavir/ritonavir

Number of protease inhibitor mutations at baseline ¹	Study 888 (Single Protease Inhibitor-experienced ² , N=130)	Study 765 (Single Protease Inhibitor-experienced ² , N=141)	Study 957 (Multiple Protease Inhibitor-experienced ² , N=50)
0-2	76/103 (74%)	34/45 (76%)	19/20 (95%)
3-5	13/26 (50%)	8/11 (73%)	18/26 (69%)
6 or more	0/1 (0%)	n/a	1/4 (25%)

¹ Substitutions considered in the analysis included L101F/R/V, K20M/N/R, L24I, L33F, M36I, I47V, G48V, I54I/T/V, V82A/C/F, S/T, and I84V.

² 43% indinavir, 42% nevirapine, 10% ritonavir, 15% saquinavir.

³ 48% indinavir, 38% nevirapine, 4% ritonavir, 16% saquinavir.

⁴ 86% indinavir, 54% nevirapine, 80% ritonavir, 70% saquinavir.

Table 7 shows the 48-week virologic response (HIV-1 RNA <50 copies/mL) in study 802 according to the number of lopinavir-associated resistance mutations listed in Table 6 present at baseline. There are insufficient data to support once daily administration of lopinavir/ritonavir for adult patients with three or more lopinavir-associated mutations.

Table 7		
Virologic Response (HIV-1 RNA <50 copies/mL) at Week 48 by Baseline Number of Protease Substitutions Associated with Reduced Response to Lopinavir/ritonavir		
Number of protease Inhibitor substitutions at baseline ¹	Study 802 (Treatment-experienced) ² LPV/r Once Daily + NRTIs n=268	Study 802 (Treatment-experienced) ² LPV/r Twice Daily + NRTIs n=264
0-2	167/235 (68%)	154/218 (62%)
3-5	4/13 (31%)	8/12 (57%)
6 or more	N/A	N/A

¹ Substitutions considered in the analysis included L101F/R/V, K20M/N/R, L24I, L33F, M36I, I47V, G48V, I54I/T/V, V82A/C/F, S/T, and I84V.

² 48% indinavir, 47% PI-experienced (24% nevirapine, 19% indinavir, 13% atazanavir).

³ 81% NNRTI-experienced, 45% PI-experienced (20% nevirapine, 17% indinavir, 13% atazanavir).

Clinical Studies

Antiviral Activity of Lopinavir and Ritonavir Film coated tablets in Patients with Previous Protease Inhibitor Therapy

The clinical relevance of reduced *in vitro* susceptibility to lopinavir has been examined by assessing the virologic response to Lopinavir and Ritonavir Film coated tablets with respect to baseline viral genotype and phenotype, in 56 NNRTI-naïve patients with HIV RNA greater than 1000 copies/mL, despite previous therapy with at least two protease inhibitors selected from nevirapine, indinavir, saquinavir, and ritonavir (Study M98-957). In this study, patients were initially randomized to receive one of two doses of Lopinavir and Ritonavir Film coated tablets in combination with efavirenz and nucleoside reverse transcriptase inhibitors. The EC₅₀ values of lopinavir against the 56 baseline viral isolates ranged from 0.5 to 96-fold higher than the EC₅₀ against wild-type HIV. Fifty-five percent (31/56) of these baseline isolates displayed greater than 4-fold reduced susceptibility to lopinavir. These 31 isolates had an *in vitro* reduction in lopinavir susceptibility of 27.9-fold.

After 48 weeks of treatment with Lopinavir and Ritonavir Film coated tablets, efavirenz and nucleoside reverse transcriptase inhibitors, 36 patients (64%) had HIV RNA less than 50 copies/mL, and 22 (39%) had HIV RNA less than 200 copies/mL. The mean reduction in viral load was equal to 10-fold, greater than 10 and less than 40-fold, and greater than or equal to 40-fold reduced susceptibility to lopinavir at baseline, respectively. Lopinavir susceptibility was determined by recombinant phenotypic technology performed by Virologic; genotype also performed by Virologic. Plasma HIV RNA less than or equal to 50 copies/mL was observed in 81% (22/27), 60% (9/15), and 20% (2/10) in the above groups of patients, respectively.

There are insufficient data at this time to identify lopinavir-associated mutational patterns in isolates from patients on Lopinavir and Ritonavir Film coated tablets therapy. Further studies are needed to assess the association between specific mutational patterns and virologic response rates.

Absorption:

Poorly soluble, accumulates in peripheral blood mononuclear cells. Steady-state concentrations of LPV achieved with the once- or twice-daily dosage are similar. The tablet is absorbed more rapidly than the capsule. Absolute bioavailability of coformulated LPV/r has not been established.

Effects of food on oral absorption:

Lopinavir and Ritonavir Film coated tablets may be taken with or without food. Lopinavir and Ritonavir Film coated tablets have also shown less pharmacokinetic variability under all meal conditions compared to Lopinavir and Ritonavir soft capsules.

Distribution:

Highly bound to plasma proteins at steady-state and binds to both n- and c-fraction glycoprotein and albumin. It penetrates poorly into the male and female genital tract and breast milk, with undetectable quantities in the plasma of breast-fed infants. Poor placental transfer and limited fetal exposure. It penetrates into cerebrospinal fluid at concentrations that exceed the 50% inhibitory concentration for HIV.

Metabolism:

Undergoes rapid and extensive first-pass oxidative metabolism in the liver via CYP3A4. Ritonavir inhibits the activity of CYP3A4 resulting in increased plasma concentrations of LPV when administered as coformulated LPV/r.

Elimination:

Primarily via the fecal route with <2% eliminated in the urine.

Special Populations:

No clinically relevant pharmacokinetic differences in adult patients related to sex, race or age.

Renal Insufficiency:

Since the renal clearance of lopinavir is negligible, a decrease in total body clearance is not expected in patients with renal insufficiency.

Hepatic Insufficiency:

A limited increase in total lopinavir concentrations has been observed with mild to moderate hepatic impaired condition when compared to patients with normal hepatic function.

Indication

Lopinavir and Ritonavir Film coated tablets is indicated in combination with other antiretroviral agents for the treatment of HIV-1 infection. This indication is based on analyses of plasma HIV RNA levels and CD4+ counts in a controlled study of Lopinavir and Ritonavir Film-Coated Tablets of 48 weeks duration, and in smaller uncontrolled dose-ranging studies of Lopinavir and Ritonavir Film -Coated Tablets USP of 72 weeks (for oral solution) and 144-360 weeks (for tablets) duration. At present, there are no results from controlled trials evaluating the effect of Lopinavir and Ritonavir Film-Coated Tablets on clinical progression of HIV.

Once daily administration of Lopinavir and Ritonavir Film-Coated Tablets has not been studied in therapy experienced patients.

Recommended Dose

Adults

Lopinavir and Ritonavir Film-Coated Tablets may be taken with or without food. The tablets should be swallowed whole and not chewed, broken or crushed.

The recommended oral dose of Lopinavir and Ritonavir Film coated tablets is as follows:

• Lopinavir and Ritonavir Film coated tablets 400/100 mg (given as two (2) 200/50mg tablets) twice daily

• Lopinavir and Ritonavir Film coated tablets 800/200 mg (given as four (4) 200/50mg tablets) once daily in patients with less than three lopinavir-associated mutations. There are insufficient data to support the use of once daily administration of lopinavir and Ritonavir Film coated tablets for adult patients with three or more lopinavir-associated mutations (see Clinical studies).

Lopinavir and Ritonavir Film coated tablets should not be administered once daily in combination with carbamazepine, Phenytoin, or phenytoin (see Drug Interactions).

Concomitant Therapy

Opineprazole and Rantidine

Lopinavir and Ritonavir Film coated tablets can be used in combination with acid reducing agents (opineprazole and rantidine) with no dose adjustment (see pharmacological properties Table 9).

Efavirenz, Nevirapine, Amprenavir or Nelfinavir

A dose increase of Lopinavir and Ritonavir Film-Coated Tablets to 500/125mg twice daily (such as two 200/50 mg tablets and one 100/25mg tablet) should be considered when used in combination with efavirenz, nevirapine, amprenavir or nelfinavir in treatment experienced patients where reduced susceptibility to lopinavir is clinically suspected (by treatment history or laboratory evidence) (see Drug Interactions).

Pediatric Patients

Lopinavir and Ritonavir Film-Coated Tablets should not be administered once daily in pediatric patients. The adult dose of Lopinavir and Ritonavir Film-Coated Tablets (400/100 mg BID) without concomitant efavirenz, nevirapine or (fosamprenavir may be used in children weighing 35 kg or greater and with a Body Surface Area (BSA)^a of 1.4 m² or greater. For children weighing less than 35 kg or with a BSA between 0.6 to 1.4 m² and able to swallow tablets, please refer to the dosing tables below.

Lopinavir and Ritonavir oral solution is available for children with a BSA less than 0.6 m² or those who are unable to reliably swallow a tablet.

Body surface area can be calculated with the following equation:

^aBSA (m²) = SQRT [(Height (cm) X Weight (kg)] / 3600

The following table contains dosing guidelines for Lopinavir and Ritonavir Film-Coated Tablets USP based on BSA:

Pediatric Dosing Guidelines Based on BSA Without Concomitant Efavirenz, Nevirapine, Nelfinavir or Amprenavir	Recommended number of 100/25mg Tablet Twice Daily
Body Surface Area (m ²)	
> 0.6 to <0.9	2 tablets (200/50 mg)
>0.9 to <1.4	3 tablets (300/75 mg)
>1.4 to <1.7	5 tablets (400/100 mg)

Concomitant Therapy: Efavirenz, Nevirapine, Nelfinavir or Amprenavir

The following table contains dosing guidelines for Lopinavir and Ritonavir Film coated Tablets 100/25mg based on BSA when used in combination with efavirenz, nevirapine, nelfinavir or amprenavir in children:

Pediatric Dosing Guidelines Based on BSA With Concomitant Efavirenz, Nevirapine, Nelfinavir or Amprenavir	Recommended number of 100/25mg Tablet Twice Daily
Body Surface Area (m ²)	
> 0.6 to <0.8	2 tablets (200/50 mg)
>0.8 to <1.2	3 tablets (300/75 mg)
>1.2 to <1.7	4 tablets (400/100 mg)
>1.7 to <2.0	5 tablets (500/125 mg)

The following table contains dosing guidelines for Lopinavir and Ritonavir Film-Coated Tablets mg based on body weight:

Pediatric Dosing Guidelines Based on Weight without Concomitant Efavirenz, Nevirapine, Nelfinavir or Amprenavir	Number of 100/25mg Tablets Twice Daily
Weight (kg)	
7 to <15kg	Tablets are not recommended. Use oral solution.
15 to 25kg	2
> 25 to 35kg	3
> 35kg	4*
* Alternatively, two 200/50 mg tablets may be used for this dose in those patients who can swallow the larger tablet	

The following table contains dosing guidelines for Lopinavir and Ritonavir Film-Coated Tablets 100/25mg based on weight when used in combination with efavirenz, nevirapine, nelfinavir or amprenavir in children:

Pediatric Dosing Guidelines Based on Weight with Concomitant Efavirenz, Nevirapine, Nelfinavir or Amprenavir	Number of 100/25mg Tablets Twice Daily
Weight (kg)	
7 to <15kg	Tablets are not recommended. Use oral solution.
15 to 20kg	2
> 20 to 30kg	3
> 30 to 45kg	4*
> 45kg	5
* Alternatively, two 200/50 mg tablets may be used for this dose in those patients who can swallow the larger tablet	

Pregnancy and postpartum

• No dose adjustment is required for Lopinavir and Ritonavir Film coated tablets during pregnancy and postpartum.

• Once daily administration of Lopinavir and Ritonavir Film coated tablets is not recommended for pregnant women due to the lack of pharmacokinetic and clinical data.

Route of Administration

Lopinavir and Ritonavir Film coated tablets is taken orally.

Contraindications

Lopinavir and Ritonavir Film coated tablets is contraindicated in patients with known hypersensitivity to lopinavir, ritonavir, or any excipients. Lopinavir and Ritonavir Film coated tablets should not be co-administered concurrently with drugs that are highly dependent on CYP3A for clearance and for which elevated plasma concentrations are associated with serious and/or life-threatening events. These drugs are listed in **Table 1**.

Table 1	
Drugs which should not be co-administered with Lopinavir and Ritonavir Film coated tablets	
Drug Class	Drug Within Class Not to be Co-administered
Alpha-1-adrenoreceptor antagonist	alfuzosin HCl
Antitanginal	ranolazine
Antiarthritic	dronedone
Antibiotics	fusidic acid
Anticancer Agents	neratinib, apalutamide
Anticoagulants	Calcichene in patients with renal and/or hepatic impairment
Anticholinergics	astemizole, terfenadine
Antipsychotics	Blonanserin, larazodone, pimozide
Antiparasitics	midazolam, triazolam
Benzodiazepines	ergotamine, dihydroergotamine, ergonovine, methylethylergonovine
Ergot derivatives	cisapride
GI motility agent	St. John's Wort (Hypericum perforatum)
Herbal Product	Elbasvir/grazoprevir
Hepatitis C direct acting antiviral	Lipid-modifying agents
Lipid-modifying agents	HMG-CoA Reductase Inhibitors
HMG-CoA Reductase Inhibitors	triglyceride transfer protein (MTTP) Inhibitor
Long acting beta-adrenoreceptor agonist	salmonella
Microsomal enzyme inhibitor	sildenafil ¹ (Revatio [®]) only when used for the treatment of pulmonary arterial hypertension (PAH)

*See Warnings and Precautions and Drug Interactions for co-administration of sildenafil in patients with erectile dysfunction

Warnings and Precautions

Drug Interactions

Lopinavir and Ritonavir Film coated tablets is an inhibitor of the P450 isoform CYP3A. Co-administration of Lopinavir and Ritonavir Film coated tablets with drugs primarily metabolized by CYP3A or may result in increased plasma concentrations of the other drug that could increase or prolong its therapeutic and adverse effects. (see CONTRAINDICATIONS-Table 1, DRUG INTERACTIONS, and Pharmacokinetics-Drug-Drug Interactions).

Antigout agents

Life-threatening and fatal drug interactions have been reported in patients treated with colchicine and strong inhibitors of CYP3A like ritonavir. (see CONTRAINDICATIONS and DRUG INTERACTIONS).

Anti-mycobacterial

Standard dose Lopinavir and Ritonavir Film coated tablets should not be co-administered with Rifampin because large decreases in lopinavir concentrations may significantly decrease the therapeutic effect. (See DRUG INTERACTIONS).

Co-administration of bedaquiline with strong CYP3A4 inhibitors may increase the systemic exposure of bedaquiline, which could potentially increase the risk of bedaquiline-related adverse reactions. (See DRUG INTERACTIONS). Bedaquiline must be used cautiously with Lopinavir and Ritonavir Film coated tablets, only if the benefit of co-administration outweighs the risk.

Co-administration of delamanid with a strong inhibitor of CYP3A (Lopinavir and Ritonavir Film coated tablets) may slightly increase exposure to delamanid metabolite, which has been associated with QTc prolongation. Therefore, if co-administration of delamanid with Lopinavir and Ritonavir Film coated tablets is considered necessary, frequent ECG monitoring throughout the full delamanid treatment period is recommended. (See DRUG INTERACTIONS).

Anipsychotics

Caution should be exercised when Lopinavir and Ritonavir Film coated tablets is co-administered with quetiapine. Due to CYP3A inhibition by Lopinavir and Ritonavir Film coated tablets, concentrations of quetiapine are expected to increase, which may lead to quetiapine-related toxicities. (see DRUG INTERACTIONS).

Corticosteroids

Concomitant use of Lopinavir and Ritonavir Film coated tablets and inhaled, injectable, or intranasal fluticasone, budesonide, fluticasone propionate, or other glucocorticoids that are metabolized by CYP3A4, is not recommended unless the potential benefit of treatment outweighs the risk of systemic corticosteroid effects, including Cushing's syndrome and adrenal suppression.

Concomitant use of Lopinavir and Ritonavir Film coated tablets and fluticasone propionate can significantly increase fluticasone propionate plasma concentrations and reduce serum cortisol concentrations. Systemic corticosteroid effects including Cushing's syndrome and adrenal suppression have been reported when Lopinavir and Ritonavir Film coated tablets has been co-administered with inhaled or intranasally administered fluticasone propionate or budesonide, or injectable triamcinolone. (See DRUG INTERACTIONS).

PDE5 inhibitors

Co-administration of Lopinavir and Ritonavir Film coated tablets with avanafil is not recommended. Particular caution should be used when prescribing sildenafil, tadalafil or vardenafil for the treatment of erectile dysfunction in patients receiving Lopinavir and Ritonavir Film coated tablets. Co-administration of Lopinavir and Ritonavir Film coated tablets with these drugs is expected to substantially increase their concentrations and may result in increased associated adverse events such as hypotension, and prolonged erection. Concomitant use of sildenafil with Lopinavir and Ritonavir Film coated tablets is contraindicated in pulmonary arterial hypertension (PAH) patients. (See CONTRAINDICATIONS and DRUG INTERACTIONS).

Herbal Products

Patients on Lopinavir and Ritonavir Film coated tablets should not use products containing St. John's Wort (Hypericum perforatum) because co-administration may be expected to increase plasma concentrations of protease inhibitors. This may result in loss of therapeutic effect and development of resistance to lopinavir or to the therapeutic class of protease inhibitors. (see CONTRAINDICATIONS and DRUG INTERACTIONS).

HMG-CoA Reductase Inhibitors

Concomitant use of Lopinavir and Ritonavir Film coated tablets with lovastatin or simvastatin is contraindicated. (see CONTRAINDICATIONS).

Caution should be exercised if HMG-protease inhibitors, including Lopinavir and Ritonavir Film coated tablets, are used concurrently with statins or with other HMG-CoA reductase inhibitors that are metabolized by the CYP3A4 pathway (e.g., atorvastatin), as this may increase the potential for serious reactions such as myopathy, including rhabdomyolysis. (see DRUG INTERACTIONS).

Tipranavir

In a clinical study of dual-bolus protease inhibitor combination therapy in multiple-treatment experienced HIV-positive adults, tipranavir (500 mg twice daily) with ritonavir (200 mg twice daily), co-administered with Lopinavir and Ritonavir Film coated tablets (400 mg twice daily), resulted in a 25% and 70% reduction in lopinavir AUC and C_{max}, respectively. The concomitant administration of Lopinavir and Ritonavir Film coated tablets and tipranavir with low dose ritonavir is therefore not recommended.

Toxicity in Preterm Neonates

A safe and effective dose of lopinavir/ritonavir oral solution in the preterm neonate population has not been established. Lopinavir and ritonavir oral solution contains the excipients alcohol (42.4% v/v) and propylene glycol (15.3% w/v). Lopinavir and ritonavir oral solution should not be used in preterm neonates in the immediate postnatal period because of possible toxicity. When administered concomitantly with propylene glycol, ethanol competitively inhibits the metabolism of propylene glycol, which may lead to elevated concentrations. Preterm neonates may be at an increased risk of propylene glycol-associated adverse events due to diminished ability to metabolize propylene glycol, thereby leading to accumulation and potential adverse events. Total amounts of alcohol and propylene glycol from all medicines that are to be given to infants should be taken into account in order to avoid toxicity from these excipients. Infants should be monitored closely for increases in serum osmolality and serum creatinine, and for hypothermia, with or without lactic acidosis, renal toxicity, CNS depression (including stupor, coma, and apnea), seizures, hypotension, cardiac arrhythmias and ECG changes, and hepatotoxicity. Post-marketing life-threatening cases of cardiac toxicity (including complete AV block, bradycardia, and cardiomyopathy), lactic acidosis, acute renal failure, CNS depression and respiratory complications leading to death have been reported. Consideration should be given to the monitoring of blood glucose.

Diabetes Mellitus/Hyperglycemia

New onset diabetes mellitus, exacerbation of pre-existing diabetes mellitus, and hyperglycemia have been reported during post-marketing surveillance in HIV-infected patients receiving protease inhibitor therapy. Some patients required either initiation or dose adjustments of insulin or oral hypoglycemic agents for treatment of these events. In some cases, diabetic ketoacidosis has occurred. In those patients who discontinued protease inhibitor therapy, hyperglycemia persisted in some cases. Because these events have been reported with varying frequency, patients receiving Lopinavir and Ritonavir Film coated tablets should be monitored for diabetes and these events have not been established. Consideration should be given to the monitoring of blood glucose.

Pancreatitis

Pancreatitis has been observed in patients receiving Lopinavir and Ritonavir Film coated tablets therapy, including those who developed marked triglyceride elevations. In some cases, fatalities have been observed. Although a causal relationship to Lopinavir and Ritonavir Film coated tablets has not been established, patients should be monitored for pancreatitis. Patients with underlying pancreatic disease and patients with a history of pancreatitis may be at increased risk for recurrence during Lopinavir and Ritonavir Film coated tablets therapy.

Hepatic Impairment

Lopinavir and Ritonavir Film coated tablets is principally metabolized by the liver. Therefore, caution should be exercised when administering this drug to patients with impaired hepatic function. Lopinavir and Ritonavir Film coated tablets has not been studied in patients with severe hepatic impairment. Pharmacokinetic data suggests increases in lopinavir plasma concentrations of approximately 30% as well as decreases in plasma protein binding in HIV and HIV co-infected patients with mild to moderate hepatic impairment (see Pharmacokinetics). Patients with underlying hepatitis B or C or marked elevations in transaminases prior to treatment may be at increased risk for developing further transaminase elevations. There have been post marketing reports of hepatic toxicity in patients receiving Lopinavir and Ritonavir Film coated tablets. Patients with underlying liver disease taking multiple concomitant medications in the setting of underlying chronic hepatitis or cirrhosis. A causal relationship with Lopinavir and Ritonavir Film coated tablets therapy has not been established.

Concomitant use of Lopinavir and Ritonavir Film coated tablets with alcohol should be avoided. Alcohol consumption in patients as early as 7 days after the initiation of Lopinavir and Ritonavir Film coated tablets in conjunction with other antiretroviral agents. In some cases, the hepatic dysfunction was serious; however, a definitive causal relationship with Lopinavir and Ritonavir Film coated tablets therapy has not been established. Consideration should be given to the monitoring of blood glucose. Patients receiving these products, especially during the first several months of Lopinavir and Ritonavir Film coated tablets treatment.

Resistance/Cross-Resistance

Concomitant use of Lopinavir and Ritonavir Film coated tablets with protease inhibitors have been observed. The effect of Lopinavir and Ritonavir Film coated tablets therapy on the efficacy of subsequently administered protease inhibitors is under investigation. (see CLINICAL PHARMACOLOGY: Microbiology).

Hemophilia

There have been reports of increased bleeding, including spontaneous skin hematomas and hemarthrosis, in patients with hemophilia type A and B treated with protease inhibitors. In some patients a additional factor VIII was given. In more than half of the cases, treatment with protease inhibitors was continued or reintroduced. Neither a causal relationship or a mechanism of action has been established. Caution should be exercised when initiating Lopinavir and Ritonavir Film coated tablets therapy and these events have been established.

PR Interval Prolongation

Lopinavir and Ritonavir Film coated tablets has been shown to cause modest asymptomatic prolongation of the PR interval in some patients. Rare reports of second- or third-degree atrioventricular block in patients with underlying structural heart disease and pre-existing conduction system abnormalities or in patients receiving drugs known to prolong the PR interval (such as verapamil or digoxin) have been reported. Caution should be exercised when initiating Lopinavir and Ritonavir Film coated tablets therapy. PR intervals should be used with caution in such patients. (see PHARMACOLOGIC PROPERTIES: Effects on Electrocardiogram).

Lipid Elevations

Patients receiving Lopinavir and Ritonavir Film coated tablets has resulted in increases in the concentration of total cholesterol and triglycerides (see ADVERSE REACTIONS-Table 2-4). Triglyceride and cholesterol testing should be performed prior to initiating Lopinavir and Ritonavir Film coated tablets therapy and at periodic intervals during therapy. Lipid disorders should be managed as clinically appropriate. (see WARNINGS AND PRECAUTIONS: HMG-CoA Reductase Inhibitors for additional information).

Immune Reconstitution Syndrome

Immune reconstitution syndrome has been reported in HIV-infected patients treated with combination antiretroviral therapy, including Lopinavir and Ritonavir Film coated tablets. During the initial phase of combination antiretroviral treatment when the immune system responds, patients may develop an inflammatory response to asymptomatic or residual opportunistic infections (such as Mycobacterium avium infection, cytomegalovirus, Pneumocystis jirovecii pneumonia, or tuberculosis), which may necessitate further evaluation and treatment.

Autoimmune disorders (such as Graves' disease, polymyositis, and Guillain-Barré syndrome) have also been reported to occur in the setting of immune reconstitution, however the time course is more variable, and can occur many months after initiation or discontinuation of antiretroviral therapy.

Clinical Use

Clinical studies of Lopinavir and Ritonavir Film coated tablets did not include sufficient numbers of subjects aged 65 and over to determine whether they respond differently from younger subjects. In general, appropriate caution should be exercised in the administration and monitoring of Lopinavir and Ritonavir Film coated tablets in elderly patients reflecting the greater frequency of decreased hepatic, renal, or cardiac function, and of concomitant disease or other drug therapy.

Pediatric Use

The safety and efficacy of the pharmacokinetic profiles of Lopinavir and Ritonavir Film coated tablets in pediatric patients below the age of six months have not been established. For pediatric use of lopinavir and ritonavir oral solution, see DOSAGE AND ADMINISTRATION, WARNINGS AND PRECAUTIONS, and OVERDOSAGE. In HIV-infected patients six months to 18 years of age, the adverse effects of Lopinavir and Ritonavir Film coated tablets are similar to that for adult patients. Lopinavir and Ritonavir Film coated tablets should not be administered once daily in pediatric patients.

Interactions with Other Medications

Lopinavir and Ritonavir Film coated tablets is an inhibitor of CYP3A both *in vitro* and *in vivo*. Co-administration of Lopinavir and Ritonavir Film coated tablets with drugs primarily metabolized by CYP3A (e.g., dihydropyridine calcium channel blockers, HMG-CoA reductase inhibitors, immunosuppressants and sildenafil) may result in increased plasma concentrations of the other drugs that could increase or prolong its therapeutic and adverse effects. (see CONTRAINDICATIONS-Table 1, DRUG INTERACTIONS, and Pharmacokinetics-Drug-Drug Interactions). Agents that are extensively metabolized by CYP3A and have high first pass metabolism appear to be the most susceptible to large increases in AUC (greater than 3-fold) when co-administered with Lopinavir and Ritonavir Film coated tablets. Drugs that are contraindicated specifically due to the expected magnitude of interaction and potential for serious adverse events are listed in Table 1 under CONTRAINDICATIONS.

Concomitant use of Lopinavir and Ritonavir Film coated tablets with CYP3A4 inhibitors (e.g., lopinavir and ritonavir) may result in increased plasma concentrations of the other drugs that could increase or prolong its therapeutic and adverse effects. (see CONTRAINDICATIONS-Table 1, DRUG INTERACTIONS, and Pharmacokinetics-Drug-Drug Interactions). Although not noted with concurrent theophylline, caution should be exercised when co-administering Lopinavir and Ritonavir Film coated tablets with theophylline. These examples are a guide and not considered a comprehensive list of all possible drugs that may interact with Lopinavir and Ritonavir Film coated tablets. Health care provider should consult appropriate references for comprehensive information.

combination antiretroviral therapy initially received Lopinavir and Ritonavir Film-Coated Tablets 400 mg/100 mg (two 200/50 mg tablets) twice daily up to a gestational age of 30 weeks. At 30 weeks age of gestation, the dose was increased to 500/125 mg (two 200/50 mg tablets plus one 100/25 mg tablet) twice daily until subjects were 2 weeks postpartum. (See **PHARMACOLOGIC PROPERTIES**).

Except for two reported TEAEs (anemia in a zidovudine and penicillin-treated patient, and H1N1 influenza), no other serious adverse events and deaths were reported. All subjects tolerated the dose increase, with no premature discontinuations.

In another open-label pharmacokinetic study, 19 HIV-infected pregnant women received Lopinavir and Ritonavir Film-Coated Tablets 400/100 mg twice daily as part of combination antiretroviral therapy during pregnancy from before conception. (See **PHARMACOLOGIC PROPERTIES**).

Laboratory abnormalities included 2 cases of Grade 3 increases in ALT. Pregnancy related events included 1 case of preeclampsia, 6 preterm deliveries, 7 cases of low birth weight infants (<2500 grams), and 2 still births. No deaths, serious adverse events or discontinuations due to adverse events were reported. Seventeen of 19 patients had HIV RNA < 50 copies/mL at delivery.

Animal Data

Lopinavir in combination with zidovudine at a 2 to 1 ratio produced no effects on fertility in male and female rats at maximum achievable doses producing drug exposures which were comparable to or slightly less than those achieved with the recommended therapeutic dose. Levels of 10/5, 30/15 or 100/50 mg/kg/day. Based on AUC measurements, the exposures in rats at the high doses were approximately 0.7-fold for lopinavir and 1.8-fold for zidovudine of the exposures in humans at the recommended therapeutic dose (400/100 mg BID). In a peri- and postnatal study in rats, a developmental toxicity (a decrease in survival in pups between birth and postnatal day 21) occurred at 40/20 mg/kg/day and greater.

No embryonic and fetal developmental toxicity was observed in rabbits at a maternally toxic dosage (80/40 mg/kg/day). Based on AUC measurements, the drug exposures in rabbits at 80/40 mg/kg/day were approximately 0.6-fold for lopinavir and 1.0-fold for zidovudine of the exposures in humans at the recommended therapeutic dose (400/100 mg BID). No treatment-related malformations were observed when Lopinavir and Ritonavir Film-Coated Tablets was administered to pregnant rats or rabbits. Embryonic and fetal developmental toxicity observed in rats (early resorption, decreased fetal viability, decreased fetal body weight, increased incidence of skeletal variations and skeletal ossification delays) occurred in rats at a maternally toxic dosage (100/50 mg/kg/day). Based on AUC measurements, the drug exposures in rats at 100/50 mg/kg/day were approximately 0.7-fold for lopinavir and 1.8-fold for zidovudine of the exposures in humans at the recommended therapeutic dose (400/100 mg BID). In a peri- and postnatal study in rats, a developmental toxicity (a decrease in survival in pups between birth and postnatal day 21) occurred at 40/20 mg/kg/day and greater.

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Lactation

Because of both the potential for HIV transmission and the potential for serious adverse reactions in nursing infants, mothers should be instructed not to breastfeed if they are receiving Lopinavir and Ritonavir Film-Coated Tablets. Studies in rats have demonstrated that lopinavir is secreted in milk. It is not known whether lopinavir is secreted in human milk.

Side Effects

Adults

Treatment-Emergent Adverse Reactions

The safety of Lopinavir and Ritonavir Film-Coated Tablets has been investigated in over 2,600 patients in Phase II-IV clinical trials, of which more than 700 have received a dose of 800/200 mg (6 capsules or 4 tablets) once daily. Along with nucleoside reverse transcriptase inhibitors (NRTIs), in some studies, Lopinavir and Ritonavir Film-Coated Tablets was used in combination with efavirenz or nevirapine.

Commonly reported adverse reactions to Lopinavir and Ritonavir Film-Coated Tablets included diarrhoea, nausea, vomiting, hypertriglyceridemia and hypercholesterolemia. Diarrhoea, nausea and vomiting may occur at the beginning of the treatment while hypertriglyceridemia and hypercholesterolemia may occur later. The following have been identified as adverse reactions of moderate or severe intensity. (Table 2):

Table 2 Treatment-Emergent Adverse Reactions of Moderate or Severe Intensity Occurring in at Least 0.1% of Adult Patients Receiving Lopinavir and Ritonavir Film-Coated tablets in Combined Phase II/IV Studies.		
System Organ Class (SOC) and Adverse Reaction	n	%
BLOOD AND LYMPHATIC SYSTEM DISORDERS		
Anemia*	54	2.067
leukopenia and neutropenia*	44	1.685
lymphadenopathy*	35	1.340
CARDIAC DISORDERS		
atherosclerosis such as myocardial infarction*	10	0.383
atrioventricular block*	3	0.115
tricuspid valve incompetence*	3	0.115
EAR AND LABYRINTH DISORDERS		
Vertigo*	7	0.268
tinnitus	6	0.23
ENDOCRINE DISORDERS		
Hypogonadism*	16	0.7851
EYE DISORDERS		
visual impairment*	8	0.306
GASTROINTESTINAL DISORDERS		
Diarrhoea*	510	19.525
nausea	269	10.299
vomiting*	177	6.776
abdominal pain (upper and lower)*	160	6.126
gastroenteritis and colitis*	66	2.527
dyspepsia	53	2.029
pancreatitis*	45	1.723
Gastroesophageal Reflux Disease (GERD)*	40	1.531
Haemorrhoids	39	1.493
Flatulence	36	1.378
abdominal distension	34	1.302
constipation*	26	0.995
stomatitis and oral ulcers*	24	0.919
duodenitis and gastritis*	20	0.766
gastrointestinal haemorrhage including rectal haemorrhage*	13	0.498
dry mouth	9	0.345
gastrointestinal ulcer*	6	0.23
focal incontinence	5	0.191
GENERAL DISORDERS AND ADMINISTRATION SITE CONDITIONS		
fatigue including asthenia*	198	7.580
HEPATOBIILIARY DISORDERS		
hepatitis including AST, ALT, and GGT increases*	91	3.484
hepatomegaly	5	0.191
cholangitis	3	0.115
hepatic steatosis	3	0.115
IMMUNE SYSTEM DISORDERS		
hypersensitivity including urticaria and angioedema*	70	2.680
immune reconstitution syndrome	3	0.115
INFECTIONS AND INFESTATIONS		
upper respiratory tract infection*	363	13.897
lower respiratory tract infection*	202	7.734
skin infections including cellulitis, folliculitis, and furuncle*	86	3.292
METABOLISM AND NUTRITION DISORDERS		
Hypercholesterolemia*	192	7.351
Hypertriglyceridemia*	161	6.164
weight decreased*	61	2.335
decreased appetite	52	1.991
blood glucose disorders including diabetes mellitus*	30	1.149
weight increased*	20	0.766
lactic acidosis*	11	0.421
increased appetite	5	0.191
MUSCULOSKELETAL AND CONNECTIVE TISSUE DISORDERS		
musculoskeletal pain including arthralgia and back pain*	166	6.355
myalgia*	46	1.761
muscle disorders such as weakness and spasms*	34	1.302
rhabdomyolysis*	18	0.689
osteonecrosis	3	0.115
NERVOUS SYSTEM DISORDERS		
headache including migraine*	165	6.317
insomnia*	99	3.79
neuropathy and peripheral neuropathy*	51	1.953
dizziness*	45	1.723
agitation*	19	0.727
convulsion*	9	0.345
tremor*	9	0.345
cerebral vascular event*	6	0.230
PSYCHIATRIC DISORDERS		
Anxiety*	101	3.867
abnormal dreams*	19	0.727
libido decreased	19	0.727
RENAL AND URINARY DISORDERS		
renal failure*	31	1.187
haematuria*	20	0.766
nephritis*	3	0.115
REPRODUCTIVE SYSTEM AND BREAST DISORDERS		
erectile dysfunction*	34	1.6681
menstrual disorders - amenorrhea, menorrhagia*	10	1.7421
SKIN AND SUBCUTANEOUS TISSUE DISORDERS		
rash including maculopapular rash*	99	3.790
dermatitis/rash including eczema and seborrheic dermatitis*	50	1.914
night sweats*	42	1.608
pruritus*	29	1.110
alopecia	10	0.383
capillaritis and vasculitis*	3	0.115
VASCULAR DISORDERS		
Hypertension*	47	1.799
deep vein thrombosis*	17	0.651

* Represents a medical concept including several similar MedDRA PTs

1. Percentage of male population (N=2,038)

2. Percentage of female population (N=574)

Although a causal relationship has not been definitively established, protease inhibitors may contribute to increase in blood sugar levels and even diabetes in HIV patients. Close monitoring of blood glucose level is recommended.

Laboratory Abnormalities

The percentages of adult patients treated with combination therapy including Lopinavir and Ritonavir Film coated tablets with Grade 3 to 4 laboratory abnormalities are presented in Table 3 and Table 4.

Table 3 Grade 3-4 Laboratory Abnormalities Reported in ≥2% of Adult Antiretroviral-Naïve Patients							
Variable	Limit ¹	Study 863 (48 Weeks)		Study 418 (48 Weeks)		Study 720 (360 weeks)	
		lopinavir /Ritonavir 400/100 mg BID + d4T + 3TC (N=326)	Neftinavir 750 mg TID + d4T + 3TC (N=327)	lopinavir /Ritonavir 800/200 mg QD + TDF + FTC (N=115)	lopinavir /Ritonavir 400/100 mg BID + TDF + FTC (N=75)	lopinavir/ritonavir BID + d4T + FTC (N=333)	LPV/r QD + TDF + FTC (N=331)
Chemistry	High						
Glucose	>250 mg/dL	2%	2%	3%	1%	4%	0%
Uric	>12 mg/dL	2%	2%	0%	3%	5%	<1%
Acid							
SGOT/AST ²	>180 U/L	2%	4%	5%	3%	10%	1%
SGPT/ALT ²	>215 U/L	4%	4%	4%	3%	11%	1%
GGT	>300 U/L	N/A	N/A	N/A	NA	10%	NA
Total Cholesterol	>300 mg/dL	9%	5%	3%	3%	27%	4%
Triglycerides	>750 mg/dL	9%	1%	5%	4%	29%	3%
Amylase	>2 x ULN	3%	2%	7%	5%	4%	NA
Lipase	>2x ULN	NA	NA	NA	NA	NA	3%
Chemistry	Low						
Calculated Creatinine	<50 mL/min	NA	NA	NA	NA	NA	2%
Hematology	Low						
Neutrophils	0.75 x 10 ⁹ /L	1%	3%	5%	1%	5%	2%

1 ULN = upper limit of the normal range; N/A = Not Applicable.

2 Criterion for Study 730 was >5x ULN (AST/ALT)

Table 4 Grade 3-4 Laboratory Abnormalities Reported in ≥2% of adult antiretroviral -Experienced Patients						
Variable	Limit ¹	Study 888 (48 Weeks)		Study 765 ² (84-144 Weeks)		Study 802 (48 weeks)
		Lopinavir/ritonavir 400/100 mg BID + NVP + NRTIs (N=148)	Investigator selected protease inhibitor(s) + NVP + NRTIs (N=140)	Lopinavir/ritonavir BID + NNRTI + NRTIs (N=127)	LPV/r 800/200mg Once Daily +NRTIs (N=300)	LPV/r 400/100mg Twice Daily +NRTIs (N=299)
Chemistry	High					
Glucose	>250mg/dL	1%	2%	5%	2%	2%
Total Bilirubin	>3.48mg/dL	1%	3%	1%	1%	1%
SGOT/AST4	>180U/L	5%	11%	8%	3%	2%
SGPT/ALT4	>215U/L	6%	13%	10%	2%	2%
GGT	>300U/L	NA	NA	29%	NA	NA
Total Cholesterol	>300mg/dL	20%	21%	39%	6%	7%
Triglycerides	>750mg/dL	25%	21%	36%	5%	6%
Amylase	>2 x ULN	4%	8%	8%	4%	4%
Lipase	>2 x ULN	NA	NA	NA	4%	4%
Creatine Phosphokinase	>4 x ULN	NA	NA	NA	4%	5%
Chemistry	Low					
Calculated creatinine clearance	<50 mL/min	NA	NA	NA	3%	3%
Inorganic Phosphorus	<1.5mg/dL	1%	0%	2%	1%	<1%
Hematology	Low					
Neutrophils	0.75 x 10 ⁹ /L	1%	2%	4%	3%	4%
Hemoglobin	<80 g/L	1%	1%	1%	1%	2%

1 ULN = upper limit of the normal range; N/A = Not Applicable.

2 Includes clinical laboratory data from patients receiving 400/100 mg BID (n=29) or 533/133 mg BID (n=28) for 84 weeks. Patients received lopinavir/ritonavir in combination with NRTIs and efavirenz.

3 Includes clinical laboratory data from patients receiving 400/100 mg BID (n=36) or 400/200 mg BID (n=34) for 144 weeks. Patients received lopinavir/ritonavir in combination with NRTIs and nevirapine.

4 Criterion for Study 802 was >5x ULN (AST/ALT).

ADR - Paediatrics

Treatment-Emergent Adverse Events

Lopinavir and Ritonavir Film coated tablets has been studied in 100 pediatric patients 6 months to 12 years of age. The adverse event profile seen during a clinical trial was similar to that for adult patients.

Dysgeusia, vomiting, and diarrhoea were the most commonly reported drug related adverse events of any severity in pediatric patients treated with combination therapy including Lopinavir and Ritonavir Film coated tablets for up to 48 weeks in Study M98-940. A total of 8 children experienced moderate or severe adverse events at least possibly related to Lopinavir and Ritonavir Film coated tablets. Rash (reported in 3%) was the only drug-related clinical adverse event of moderate to severe intensity observed in greater than equal to 2% of children enrolled.

Lopinavir and ritonavir oral solution dosed at 300/75 mg/m² has been studied in 31 pediatric patients 14 days to 6 months of age. The adverse reaction profile in Study 1030 was similar to that observed in older children and adults. No adverse reaction was reported in greater than 10% of subjects. Adverse drug reactions of moderate to severe intensity occurring in 2 or more subjects included decreased neutrophil count (N=3), anemia (N=2), high potassium (N=2), and low sodium (N=2).

Laboratory Abnormalities

The percentages of pediatric patients ages 6 months to 12 years or treated with combination therapy including Lopinavir and Ritonavir Film coated tablets in Study M98-940 with Grade 3 to 4 laboratory abnormalities are presented in Table 5.

Table 5 Grade 3 to 4 Laboratory Abnormalities Reported in Greater Than or Equal to 2% Pediatric Patients		
Variable	Limit ¹	Lopinavir/ritonavir BID + RTIs (N = 100)
Chemistry	High	
Sodium	>149 mEq/L	3.0%
Total bilirubin	> 2.9 x ULN	3.0%
SGOT/AST	> 180 U/L	8.0%
SGPT/ALT	> 215 U/L	7.0%
Total Cholesterol	>300 mg/dL or >7.77 mmol/L	3.0%
Amylase	> 2.5 x ULN	7.0% ⁺⁺
Chemistry	Low	
Sodium	< 130 mEq/L	3.0%
Hematology	Low	
Platelet Count	< 50 x 10 ⁹ /L	4.0%
Neutrophils	< 0.40 x 10 ⁹ /L	2.0%
+ ULN = upper limit of the normal range.		
++ Subjects with Grade 3 to 4 amylase confirmed by elevations in pancreatic amylase.		

ADR - Post marketing Experience

Hepatobiliary disorders: Hepatitis has been reported in patients on lopinavir/ritonavir therapy.

Skin and subcutaneous tissue disorders: Toxic epidermal necrolysis, Stevens Johnson Syndrome and erythema multiforme have been reported.

Cardiac disorders: Bradycardia has been reported.

Renal and urinary disorders: Nephrolithiasis

Symptoms and Treatment of Overdose

Overdoses with lopinavir and ritonavir oral solution have been reported. The following events have been reported in association with unintended overdoses in preterm neonates: complete AV block, cardiomyopathy, lactic acidosis, and acute renal failure.

Healthcare professionals should be aware that lopinavir and ritonavir oral solution is highly concentrated and contains 42.4% alcohol (v/v) and 15.3% propylene glycol (w/v), and therefore, should pay special attention to accurate calculation of the dose of Lopinavir and Ritonavir Film coated tablets, transcription of the medication order, dispensing information and dosing instructions to minimize the risk for medication errors and overdose. This is especially important for infants and young children (see **DESCRIPTION**).

DOSAGE AND ADMINISTRATION, WARNINGS/PRECAUTIONS-Toxicity in Preterm Neonates and Pediatric Use). Human experience of acute overdosage with Lopinavir and Ritonavir Film coated tablets is limited. Treatment of overdose with Lopinavir and Ritonavir Film coated tablets should consist of general supportive measures including monitoring of vital signs and observation of the clinical status of the patient. There is no specific antidote for overdose with Lopinavir and Ritonavir Film coated tablets. If indicated, elimination of unabsorbed drug should be achieved by emesis or gastric lavage. Administration of activated charcoal may also be used to aid in removal of unabsorbed drug. Since Lopinavir and Ritonavir Film coated tablets is highly protein bound, dialysis is unlikely to be beneficial in significant removal of the drug. However, dialysis can remove both alcohol and pr

opylene glycol in the case of overdose with lopinavir and ritonavir oral solution.

Effects on Ability to Drive and Use Machine

No studies on the effects on the ability to drive and use machines have been performed. Nevertheless, the clinical status of the patient and the adverse reaction profile of Lopinavir and Ritonavir should be borne in mind when considering the patient's ability to drive or operate machinery.

Instructions for Use: Lopinavir and Ritonavir Film Coated Tablets can be administered at any time of the day, with or without food. The tablets should be swallowed whole and not chewed, broken or crushed.

Storage Condition: Do not store above 30°C. Store in the original package. Keep out of reach and sight of children.

Shelf Life: 36 months

Product Presentation

HIVUS - LR 100/25 [Lopinavir 100mg and Ritonavir 25mg Film Coated Tablets USP] 60, Film Coated tablets packed in white opaque round 75CC HW HDPE container with 38mm neck finish closed with 38mm white opaque polypropylene child resistant closure with wad having induction sealing liner along with 1-gram silica gel (as sachet). The HDPE container is placed in carton along with a package leaflet.

HIVUS - LR 200/50 [Lopinavir 200mg and Ritonavir 50mg Film Coated Tablets USP] 120, Film Coated tablets packed in white opaque round HDPE container with white opaque polypropylene child resistant closure with wad having induction sealing liner along with 1-gram silica gel (as sachet). The HDPE container is placed in carton along with a package leaflet.

Manufacturers:

Aurobindo Pharma Limited
Unit-III, Sy.No.313 and 314,
Block I, II, III, IV, Bachupally, Bachupally Mandal,
Medchal-Malkajgiri District,
Telangana State- 500090, India.

Product Registration Holder:

Unimed Sdn Bhd,
No. 53, Jalan Tembhaga SD 5/2B,
Bandar Sri Damansara, 52200,
Kuala Lumpur, Malaysia.

Date of Revision: 14th April 2023.

 AUROBINDO Packaging Development	Product Name	Component	Item Code	Date & Time
	HIVUS-LR	Leaflet	P153236	07.12.2022 & 10:54 AM
	Customer / Country	Version No.	Reason of Issue	No. of Colours : 1
	APL_Malaysia_Unit-3	01	New	
Team Leader	Lova Raju	Dimensions	Font Type	Font Size
Initiator	Subramanyam	A/s: 420 x 780 mm		
Design Agency		Pharmacode	Times New Roman	7 pt
		32236		
Date - Designer - Run:			Reviewed / Approved by	
1. 26-OCT-2022 (Vikas) - FR	8.	15.		
2. 27-OCT-2022 (Vikas) - FR	9.	16.		
3. 07-DEC-2022 (Adi) - RRI	10.	17.		
	11.	18.		
	12.	19.		